James Ritchie

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Professional Experience

- **StretchSense** **Edinburgh, UK**
  Senior Machine Learning Engineer *April 2024 - July 2025*
  - Lead the development of machine learning models for our next-generation motion capture gloves integrating our proprietary sensor technology.
  - Built models to perform gesture recognition enabling the gloves to be used as a controller for virtual reality training applications using Python and scikit-learn.
  - Researched and implemented novel deep learning architectures for real-time hand pose estimation from sensor data with small and large datasets using PyTorch and TensorFlow, reducing error sizes by 15% compared to previous models.
  - Developed a computer vision system to gather data for training hand models.
  - Collaborated with software engineers as part of an agile team to build and deploy models that can both train and infer in real-time on the edge.
- **Amazon** **Berlin, DE (Remote)**
  Applied Science PhD Placement *September 2021 - November 2021*
  - Worked on recommender systems in the Amazon Music ML team.
  - Implemented and evaluated a new learning-to-rank approach using deep learning.
  - Passed placement review with excellent feedback and received an "inclined-to-hire" offer.
- **University of Edinburgh** **Edinburgh, UK**
  Teaching Support Provider *September 2018 - August 2022*
  - Tutoring and marking for postgraduate-level machine learning courses alongside PhD studies.
  - Nominated for the ‘Best Student Who Teaches’ award in 2019.
- **Kriya** **London, UK**
  Data Scientist *December 2015 - August 2017*
  - Created machine learning models using Python/NumPy/scikit-learn for loan pricing and marketing at a financial technology startup specialising in invoice financing.
  - Deployed models in production as microservices on AWS and Azure.
  - Worked as part of an agile team to identify and solve problems using data, communicating results to a range of stakeholders across the business.
  - Achievements included:
    - Scraping and cleaning data from the Companies House API and using it to train a machine learning model to identify potential customers, resulting in 200+ high quality sales leads.
    - Took the initiative to implement a new ETL pipeline from the company platform to our data warehouse, which increased the sync success rate from 85% to 100%.
    - Built an application for the risk team to scrape news stories about companies and perform sentiment analysis using natural language processing (NLP). This successfully flagged issues with customers two weeks faster than the credit-scoring agencies.

- **Skin Analytics** **London, UK**
  Software Engineer *July 2014 – February 2015*
  Startup using computer vision and machine learning for skin cancer diagnostics. Worked with the CTO and image processing engineers on infrastructure and apps for the company's products.

Education

- **University of Edinburgh** **Edinburgh, UK**
  PhD in Machine Learning *September 2018 - March 2023*

  Thesis: Bayesian Inference for Challenging Scientific Models
  Supervisor: Professor Iain Murray
  Researched new methods to apply Bayesian inference to complex models of scientific phenomena from a range of fields including physiology and astronomy, using recent advances in probabilistic machine learning, generative AI, deep learning and statistics. Implemented new models and designed experiments using the Python scientific stack and PyTorch that successfully validated these methods on large real-world datasets.
- **University of Edinburgh** **Edinburgh, UK**
  MSc(R) Data Science *September 2017 - August 2018*

  Result: Distinction
  Thesis: Bayesian Hyperparameter Optimisation with Progressively Larger Models
- **University of Cambridge** **Cambridge, UK**
  MEng MA Information and Computer Engineering *October 2010 - June 2014*

  Result: Honours with Merit

Open Source Software

- **scikit-rvm** *github.com/JamesRitchie/scikit-rvm*
  Implementation in Python/NumPy/SciPy using the scikit-learn API of the Relevance Vector Machine (RVM) technique used for probabilistic regression and classification in machine learning. I gave a lightning talk on this project and scikit-learn development in general at PyData London.

Skills and Technologies

Programming Languages: Python (Professional), C++ (Limited), C# (Limited).
Machine Learning/Statistics: Bayesian inference, deep learning, computer vision, data visualisation, NumPy, SciPy, scikit-learn, Pandas, PyTorch, Tensorflow, Jax, Matplotlib, GPU computing.
Software Engineering: Version control (Git), unit testing, continuous integration, web application development, cloud computing (AWS), containerisation (Docker), Linux, SQL (PostgreSQL, Amazon Redshift).